

EDWARD JENNER, F.R.S., AND THE CUCKOO

By E. L. SCOTT

IN January 1973 the 150th anniversary of the death of Edward Jenner, the originator of vaccination, was modestly commemorated in the small town of Berkeley, Gloucestershire, where he was born and lived for most of his life (1). Jenner was himself the subject of controversy during and after his lifetime, on the one hand lauded as a selfless saviour of mankind, and on the other denounced as a self-seeking charlatan; his discovery likewise has virtually never ceased to be a subject of debate (2). There is no dearth of books and articles about Jenner and vaccination (3) and the subject of the present note is an unconnected and very minor discovery of Jenner's—nevertheless one which is not only intrinsically interesting to naturalists, but also pertinent to any study of the character and integrity of its author, for the discovery was still being disputed more than a century after it was made.

Edward Jenner was the first to publish, in 1788, an accurate account of what happens to the unfortunate young of the parent bird in whose nest a cuckoo deposits one of her eggs. Yet to consult the *Dictionary of National Biography* for details of Jenner's life is to read, in an article published in 1892, that the 'absurdity' of this account had been demonstrated by the naturalist Charles Waterton. A few years earlier Charles Creighton (1847–1927), the epidemiologist and medical historian, whose intense opposition to vaccination led him to denigrate Jenner in every conceivable way, had described his paper on the cuckoo as mainly 'a tissue of inconsistencies and absurdities' (4).

The parasitic habits of the European cuckoo (*Cuculus canorus*) have long been known. Aristotle wrote that the bird in whose nest the cuckoo deposits her egg hatches it and rears its occupant, and 'as they say, . . . when the young cuckoo has grown big, thrusts her own brood out of the nest and lets them perish'. Some, however, said that the mother kills her own brood and 'gives them to the alien to devour, despising her own young owing to the beauty of the cuckoo'. Others again said that the legitimate occupants of the nest were devoured by the mother cuckoo, or that they simply died of starvation, all the food being taken by the young cuckoo—some even accused the latter of killing the other nestlings by its superior strength (5).

It is not known what first prompted Jenner's interest in the cuckoo, but from a reference to it in the first known letter from John Hunter to Jenner,

written in June 1773 (6) (unfortunately none of Jenner's letters to Hunter have survived), it seems that he began his observations shortly after returning to Berkeley from London, where he had been Hunter's house-pupil. Not, however, until early in 1787 did Jenner send to Hunter a paper on the cuckoo which he hoped would be read to the Royal Society—and, incidentally, further his own prospects of election.

In this paper, the manuscript of which has survived (7) Jenner wrote: 'Soon after the Hedge Sparrow has disengaged the young Cuckoo and some of her own offspring from the shell, she singles out all her own young ones and throws them out of the nest; and with them the eggs, tho' containing living birds that remain unhatched . . . all are left to perish together, either entangled about the bush which contained the nest, or on the ground under it'. This passage was, however, the result of conjecture based on the observation implied in the concluding phrases. The paper was read on 29 March 1787 (8) and in the following June he found he had been wrong, for he saw a newly-hatched cuckoo actually ejecting its fellow nestlings. He requested that his paper as it stood should not be published (9), then revised it; and the altered paper (or part of it) was read on 13 March 1788 (10). The observation which led to the revision is graphically described in the published paper (11):

The little animal, with the assistance of its rump and wings, contrived to get the bird upon its back, and making a lodgement for the burden by elevating its elbows, clambered backward with it up the side of the nest till it reached the top, where resting for a moment, it threw off its load with a jerk, and quite disengaged it from the nest.

That this was met by some incredulity is shown by a letter from Hunter to Jenner shortly after this second reading—apparently in reply to an enquiry about his election to the Society. Hunter said that Sir Joseph Banks had thought that this should be delayed until the paper had been printed, so as to 'let the people rest a little upon it,—for he says there are many who can hardly believe it wholly. . . ' (12).

Creighton wrote that Jenner's account 'had been accepted by all ornithologists' (4) and cited Thomas Pennant (one of Gilbert White's correspondents (13)), who in his *British Zoology* had originally given the 'sensible explanation' (14) that the young cuckoo destroyed the genuine offspring by overlaying them (15) but altered this in the next edition (16).

Jenner's observation was acknowledged and accepted by George Montagu (1751–1815) (17) and John Latham (1740–1837) (18), but John Gould (1804–1881), in his monumental *Birds of Great Britain*, expressed his disbelief (19). He

said there was much diversity of opinion as to how the young birds were ejected: 'the general belief is that they are shouldered out by the parasite; but from this I entirely dissent, for the simple reason that, judging from my own observation, I do not believe that at *the end of the third day* the parasite has the physical power to eject the rightful possessors from a deep cup-shaped or a domed nest' (20).

One is therefore surprised to see, opposite this statement, a coloured lithograph depicting a young cuckoo ejecting the nestlings from the nest of a meadow pipit, while the parent bird sits by the nest apparently unmoved. The explanation is to be sought in the introduction, where the birds are arranged systematically, by order, family and genus. Under the entry for the cuckoo Gould wrote: 'I now find that the opinion ventured in my account of this species as to the impossibility of the young Cuckoo ejecting the young of its foster-parents at the early age of three or four days is erroneous, for a lady of undoubted veracity and commendable ability as an observer of nature and as an artist has actually seen the act performed, and has illustrated her statement of the fact by a sketch taken at the time, a tracing of which has been kindly sent to me. . .' (21).

It seems therefore particularly unfortunate that the *D.N.B.* article referred to earlier should have been written by someone who had been considerably influenced by the naturalist and traveller Charles Waterton (1782–1865). This was Sir Norman Moore, president of the Royal College of Physicians from 1918 to 1922, who wrote over 450 articles for the *Dictionary* and had personal knowledge of Waterton, whom he greatly admired. A few years after Waterton's death Moore edited an edition of the *Essays of a Naturalist*, which included a life of the author and an appendix of his letters.

Early in 1836 Waterton sent an article on the habits of the jay to J. C. Loudon, for inclusion in his *Magazine of Natural History*. In a reference to the young of the jay being born blind, he remarked on the differences, in degrees of helplessness, between young birds born blind and those with their eyes open. An American friend, he said, had asked his opinion of 'our English account' (Jenner's name was not mentioned) of 'a young cuckoo, which, on the very day that it was hatched, was actually seen retrograding up the side of a hedge sparrow's nest with a young hedge sparrow on its back. . . . No bird in the creation could perform such an astounding feat under such embarrassing circumstances. The young cuckoo cannot, by any means, support its own weight during the first day of its existence' (22). Waterton ended by saying he would rather believe the story of the young Hercules throttling two snakes in his cradle.

The American friend was George Ord, a naturalist in Philadelphia, whom Waterton had met whilst touring the United States. In a letter from Waterton to Ord, included in Moore's edition of the *Essays*, he wrote: 'You are quite right with regard to Jenner's preposterous account of the young cuckoo. He never saw what he relates' (23). Moore's article on Jenner was on the whole favourable and he clearly felt that some face-saving was necessary: 'The explanation appears to be that Jenner employed a boy, his nephew Henry, to make these observations, who, too indolent to watch, gave an imaginary report' (24). It is true that Jenner sent the boy on a daily journey of four or five miles to a farm at Clapton belonging to Deborah Hooper, Edward Jenner's aunt, to report on some nests being watched, and undoubtedly the boy found it 'tiresome' (25), but it is difficult to believe that Jenner could have given such a vivid description of something he had not himself witnessed (26).

The practical difficulty of observing the act described by Jenner, accounting to some extent for the long suspension of belief, is perhaps underlined by the fact that no photographic evidence was exhibited to the public until November 1921. A film made by the ornithologist Edgar Chance on the life of the cuckoo was then shown to members of the Zoological Society of London. Chance subsequently quoted from a correspondent who had seen the film and whose most lasting impression was the sight of a young meadow pipit being borne on the back of a young cuckoo's neck 'almost exactly as a man delivering coals carries the sack. . . . Most horrible of all was the cold-blooded unconcern of the mother Pipit. . .' (27).

NOTES

- (1) Jenner died on 26 January 1823. On 6 and 7 January 1973 a new documentary film about him was shown in the church in which he is buried and of which his father was once vicar.
- (2) It was reopened as recently as 1965. See P. E. Razzell, 'Edward Jenner: The History of a Medical Myth', *Med. Hist.*, 9, 216-229 and replies *ib.*, 381-384 (1965).
- (3) Any biography must rely ultimately on J. Baron, *The Life of Edward Jenner, M.D.*, vol. I, London, 1827 (repr. 1838), vol. II, London, 1838, though its exaggeratedly eulogistic style renders it unsatisfactory. Important recent works are W. R. le Fanu, *Bio-bibliography of Edward Jenner*, London, 1951, and C. W. Dixon, *Smallpox*, London, 1962. For a short account of Jenner's life see G. R. Cameron, *Notes and Records R. Soc. Lond.*, 7, 43-53 (1950).
- (4) C. Creighton, *Jenner and Vaccination*, London, 1889, p. 17.
- (5) *Historia Animalium, The Works of Aristotle*, vol. IV, trans. D'Arcy Wentworth Thompson, Oxford, 1910, Book IX, Ch. 29.
- (6) In J. Baron, *op. cit.*, vol. I, p. 28, undated. W. R. le Fanu dated many of the letters cited by Baron from inspection of the originals and gives the date of this letter as 13 June 1773.

- (7) Roy. Soc., Letters and Papers, Decade IX, No. 37.
- (8) Roy. Soc. Journal Book, vol. 32, 446–449 (1785–1787).
- (9) A note on the ms. reads 'Cancelled at the desire of the author, C. B.' (presumably written by Charles Blagden). See letter from Jenner to Sir Joseph Banks (undated) and reply from Banks, 7 July 1787, in Baron, *op. cit.*, I, p. 77.
- (10) Roy. Soc. Journal Book, vol. 33, 153–154 (1787–1790). The entry ('The reading of Mr. Jenner's paper . . . was concluded') implies that only part of the revised paper was read.
- (11) *Phil. Trans. R. Soc. Lond.*, 78, 219–237 (1788).
- (12) Baron, *op. cit.*, vol. I, p. 78.
- (13) White refers to the cuckoo in several of his letters, but does not mention this problem. The first edn. of *The Natural History and Antiquities of Selborne* was apparently published late in 1788 (though the t.p. says 1789—see introduction by J. Fisher to an edn. of 1947 (The Cresset Press, London), p. ix) so it is unlikely that White would have seen Jenner's paper before sending his ms. to the printers.
- (14) Creighton's words—he also denied the existence of the hollow in the back of the young cuckoo, which Jenner accurately reported, noting how it filled up after a few days, when the disposition to eject other birds ceased. The existence of this structural adaptation has since been well substantiated. Creighton apparently had no grounds for his views other than his desire to discredit Jenner.
- (15) T. Pennant, *British Zoology*, 4th edn., vol. I, London, 1776, p. 201.
- (16) *Ibid.*, the posthumous edn. of 1812, edited by his son David Pennant, vol. I, p. 308.
- (17) *Ornithological Dictionary; or Alphabetical Synopsis of British Birds*, London, 1802, 2 vols., pages unnumbered. James Rennie, who edited and enlarged the posthumous 2nd edn., relates how he verified Jenner's observation by repeatedly putting a young swallow in a nest from which a young cuckoo had ejected four young hedge sparrows. He noted, as had Jenner, that after about ten days, the young cuckoo no longer tried to eject the other bird.
- (18) *A General History of Birds*, vol. III, Winchester, 1822, p. 260.
- (19) *The Birds of Great Britain*, vol. III, London, 1873, page facing plate 68 (pages are unnumbered).
- (20) The reason for his italics becomes clear on the next page, where he says he could believe it of the cuckoo when it had gained sufficient strength, but he had observed the young birds lying on the ground below the nest within three days of the cuckoo being hatched.
- (21) The sketch, which was reproduced as a small woodcut in the introduction and obviously formed the basis of the much larger lithograph, was made by Mrs Jane Blackburn and first appeared in a small anonymous volume, *The Pipits*, Glasgow, 1872. The circumstances under which it was made were subsequently related in *Nature, Lond.*, 5, 383 (1872) and reprinted by Gould in his introduction. She implies that she had not seen Jenner's account until after her own observation.
- (22) *Mag. Nat. Hist.*, 9, 189 (1836). The essay was repr. in C. Waterton, *Essays on Natural History*, London, 1838, pp. 226–231.
- (23) *Ibid.*, ed. with a Life of the Author, by Norman Moore, London, 1871, p. 555.
- (24) *D.N.B.*, vol. 29, 321–324 (1892).
- (25) Baron, *op. cit.*, vol. I, p. 85.

- (26) A note-book covering the relevant period, in the possession of the Royal College of Physicians, was published in 1931 (*The Note-book of Edward Jenner*, ed. F. D. Drewitt, London, 1931). Though the crucial observation is not recorded, it is possible to deduce from the entries that Jenner became aware of the real culprit near the end of June 1787 (see pp. 25-27). On 28 June he observed a young cuckoo turning out not only a young hedge sparrow, but another young cuckoo hatched in the same nest. Dr Drewitt's introduction formed the basis of Ch. III of his *Life of Edward Jenner* (London, 1931). This chapter, which was extended in the 2nd edn. (1933), gives a good account of Jenner's work as a naturalist, and the present writer is indebted to it for some references—unfortunately there is no mention of Jenner's revision of his paper.
- (27) E. P. Chance, *The Cuckoo's Secret*, London, 1922, p. 212.